

Operations Practice

Increasing transparency in megaproject execution

Cost and schedule overruns are the norm in big capital projects. Here's how to uncover their causes—and tackle them.

This article is a collective effort by Daniel Ahmoye, Matthew Roberts, Alpa Saxena, Julia Slayden, and Kevin Stokvis, representing views from McKinsey's Operations Practice.



In the coming decades, a significant amount of capital—which our colleagues have estimated at \$130 trillion—is likely to be spent on projects to decarbonize the global economy and renew critical infrastructure. Few organizations are prepared to handle this capital influx with the speed and efficiency it demands, however, with many already struggling to deliver their existing capital programs. A recent survey of senior project executives found that, on average, projects overrun their budgets and schedules by 30 to 45 percent.

Why is it so hard to deliver capital projects on time and on budget? High levels of complexity are part of the problem. Executing major projects is always challenging, requiring complex, multilevel collaboration across an ecosystem of partner organizations and the coordination of resources at the front line. That complexity makes it vital that leaders have clear visibility of project performance. No big project is going to run seamlessly; organizations will need to manage all sorts of challenges, from supply chain delays to skills shortages. Fixing those problems is easiest when owners intervene quickly and decisively to address them.

Yet all too often, leaders lack a clear picture of what is happening on the ground. It can take days or weeks for the news of execution delays to filter up through the organization. And when information does arrive, it may not be the right information at the right level of detail for the recipients to understand the root cause of the execution delay. This makes it difficult for project leaders to take timely mitigation and recovery actions.

This lack of transparency is driven by multiple reporting issues. Data on progress may be siloed in disparate systems used by many different contractors. Critical data may be collected in ways that make aggregation and analysis difficult, such as nonstandardized PDF forms or handwritten notes. Teams on the ground may not be disciplined in keeping accurate and up-to-date records. Even on the same project, dozens of measures of performance may not be consistently tracked across the enterprise or standardized across

contractors. That means that inaccurate, out-of-date, or misleading information is often presented to project leaders for decision making.

Additionally, multiple stakeholders with different needs and objectives are looking for different insights from project data. Project directors need to understand how issues will affect the overall budget, for example, while project managers may be more concerned about how delays in materials supplied to the site will affect construction schedules. If they don't receive the right information at the needed frequency and granularity, it becomes even harder for owners and contractors to generate insights, identify issues and opportunities, and make decisions at the right time.

To address these challenges, organizations should focus on three key areas. First, they need *new technical solutions* to capture and analyze project data. Second, they need *improved project performance management processes* that enable rapid, effective decision making based on that data. Third, they need *project leaders with the right mindsets, behaviors, and capabilities* to support high productivity, quality, and consistency in project execution. None of these elements are optional. A more effective project management approach “recipe” requires all three ingredients, and they work together to deliver improved project performance.

Revealing the truth: Using data and analytics to transform data into insights

Complex projects generate huge amounts of data. Taken together, engineering documents, schedules, purchase orders and invoices, staff time sheets, and countless other items can paint a rich and detailed picture of a project's status. But to see that picture, management teams need a better way to access and interpret all their data.

Technical solutions involve setting up the digital infrastructure necessary to create project transparency among execution teams and top management. Historically, project information

has been captured in disparate and fragmented systems, with linkages between information that can drive differentiated insights often lost in the shuffle. Implementing the right technical system increases the speed of ingestion, aggregation, and visualization of project data from different sources.

This allows organizations to apply analytics and draw predictive insights that enable decisive actions to mitigate risks. Such insights might include early warnings or red flags when slow progress on one or more activities puts the on-time achievement of critical project milestones in doubt, or where unresolved design or supply chain issues mean other activities are not ready to commence on their scheduled dates. Management teams can also use

predictive tools to conduct what-if analyses. This might allow them to compare the likely impact of various interventions, such as productivity improvement in key tasks, or the sharing of equipment and personnel between projects.

A unified technical infrastructure (see sidebar, “Visualizing a better digital control tower”) helps the entire organization work together to keep execution on track, ensuring that project executives see a comprehensive picture, while on-the-ground leadership has all the necessary data to make decisions. For example, if the delivery of a piece of critical equipment to the job site is delayed, communicating that issue quickly to multiple stakeholders allows everyone to take the most appropriate actions. The construction supervisor sees the delay in their project delivery dashboard and decides to redeploy their crew to another critical task while they wait. Procurement or logistics sees the equipment is delayed and begins to work on expediting procedures. The executive team sees that it is delayed but recognizes that schedule slippage is expected to be minimal, based on the mitigating actions that construction and procurement have implemented.

When one chemical company faced cost overruns and delays in the construction of a large petrochemical facility, it set up a digital control tower to act as central data repository and analysis hub. The control tower helped the project team to identify, manage, and deliver more than \$75 million in cost savings over the remaining course of the project—and it found those opportunities quickly. Setting up an effective and targeted control tower can take as little as two or three weeks, with dedicated resources and clear alignment from leadership, without requiring significant additional technical infrastructure.

Taking action: Improving project management processes to drive an effective performance cadence

Good data is necessary for successful project execution, but it isn't sufficient. Projects also need management processes that enable effective

Visualizing a better digital control tower

To give leadership teams access to accurate, timely, and actionable information they need, a project data management system should have certain key capabilities:

1. It should ingest data from multiple sources, condition that data, and present it with effective visuals.
2. It should stay up to date, always using the latest, most accurate information available.
3. It should provide a single source of truth for all project stakeholders.
4. It should offer insights that support early and rapid identification of issues and opportunities.
5. It should enable progressive discovery, summarizing information at the right level for different stakeholders, while also allowing users to identify the underlying drivers to aid problem diagnosis and action planning.
6. It should be adaptable to meet changing project requirements.
7. It should be easy to use and simple to maintain.

Good data is necessary for successful project execution, but it isn't sufficient. Projects also need management processes that enable effective decision making and drive performance on the ground.

decision making and drive performance on the ground. Too often, however, project management meetings struggle to meet that ideal. Common issues include discussions based on anecdotes or qualitative estimates of progress rather than hard facts, and a tendency to look backward rather than forward, focusing on explaining what happened yesterday rather than on problem solving for better performance tomorrow. These problems can be exacerbated by the tools companies use. For example, paper reports or PowerPoint presentations don't allow leaders to drill down into project data and understand the root cause of issues. Without standard agendas and action logs, it is hard to ensure that decisions are followed through. And disconnections inside the organization can prevent the effective two-way flow of information and decisions between the top management team and the front line.

Improved project performance management processes are needed to enable teams and individuals to work toward the common goal of timely and within-budget project delivery. Such processes begin with fact-based discussions using real-time, or near-real-time data. They take an integrated view of project performance, looking across phases (for example, procurement, engineering, and construction by trade) and project areas. And they are structured to emphasize forward-looking problem solving.

In a robust and high-performing project team, information is cascaded through the management

infrastructure to enable decision making at the right levels. A strong performance management culture includes regular project meetings, defined decision processes (including escalation paths), structured performance dialogues, and dedicated physical spaces for project decision making (that is, command centers). Data is aggregated and presented at the right level for each meeting so that site supervisors and team leaders see a more focused, granular view than the senior management team, for example.

A project's digital control tower and its project performance management processes are closely linked. With instant access to accurate and comprehensive data, management teams can focus their attention on the issues that have the highest potential to affect project outcomes, rather than spending time on meandering discussions based on incomplete information. And if that data is held in the cloud, staff can access it and contribute to discussions wherever they are. Conversely, even perfect information availability won't improve project outcomes unless an organization has good systems to support decision making and drive action.

A recent large liquefied natural gas project improved its performance management systems to reduce schedule slippage. It introduced shop floor production measurement practices and developed a standardized intervention cycle that enabled the front line to improve performance on an ongoing basis. This, combined with a digital control tower as a single source of truth for project data, allowed

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the owner to collaborate more closely with the contractor and improve the project schedule.

Enabling the team: Enhancing the capabilities mindsets and behaviors of project stakeholders

A focus on organizational capability building is needed to turn the insights obtained from improved transparency into results on the ground. That requires a range of skills, including knowing how to interpret data and how to manage constructive, fact-based discussions. Project delivery personnel should know how to create, and modify, complex plans and schedules. And frontline teams need lean construction skills to unlock performance improvements through the standardization and continuous improvement of tasks.

Mindsets and behaviors are just as critical. To keep projects on track, leaders need to be activists, engaging with their teams and working with them to improve frontline performance. They need to be resilient too, able to assimilate new information quickly and adapt their plans as things change on the ground. And all project team members should be adept at problem solving, finding fast, effective solutions to issues as they arise.

A digital control tower can enable all three of these critical attributes: helping leaders focus their attention on the issues that matter most; delivering relevant information in a clear, comprehensible format; and providing both data leaders and data managers with the information they need to find optimal solutions to difficult problems. For example, one Asia-Pacific-based engineering, procurement, and construction (EPC) company with a large

project portfolio had long struggled to address data quality and performance management issues that exacerbated delays. Implementing a digital control tower allowed leaders to aggregate, integrate, and condition project data that previously had been siloed across multiple sources. The result? For the first time, structured deep dives could reliably uncover the root causes of problems, leading to faster, more accurate, and more agile decision making.

The full potential became clear when the company provided focused training for working teams on the control tower's capabilities and use, leading to its adoption in weekly progress meetings for a priority project that faced a mix of technical, engineering, and contractual issues. After quick progress in implementing solutions, the company scaled the digital control tower across a variety of projects to improve transparency across its entire portfolio.

Transparency helps reduce project risk and accelerate execution. Transparency ensures top management understands the project's progress and estimated trajectory. It enables organizations to undertake proactive performance management, anticipate risks, develop mitigation plans to avoid cost overruns, and identify areas of opportunity for acceleration and additional value capture. Owners and contractors can bring increased transparency to their projects by implementing new technical solutions, improving project performance management processes, and fostering the right mindsets, behaviors, and capabilities among their project leaders.

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